

LUMENCORE

Portfolio Review Brief | October 4, 2026

Science, geometry and imagination - connected by reproducible evidence.

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LumenCore combines implemented software, evidence tooling, forecasting research and geometric design. Its commercial focus is a bounded buyer-owned evaluation: define a consequential decision, freeze authorized inputs and an accepted baseline, compare candidates, preserve adverse results, and return an inspectable decision record.

The artwork expresses the vision. The equations define a model. Experiments determine what can be claimed.

What a reviewer can inspect

Portfolio layer	Review object	Current boundary
Evidence infrastructure	Versioned source, manifests, replay instructions, claim controls and release history	Integrity and reproducibility controls do not by themselves validate a scientific conclusion.
Forecasting and quantitative research	Frozen comparisons, coverage, uncertainty and negative findings	Historical research; buyer acceptance and independent outcomes require separate evidence.
Geometry research	Candidate registry, task-specific comparison lanes and one fresh ideal-DC screen	No universal geometry champion or physical signal-efficiency claim.
Visual design	Mathematical constructions, interactive studies and cinematic concepts	Visual detail is distinct from measured physical performance.

The geometry catalog has expanded

26 families across 11 task lanes, up from 18 families across nine lanes. Eight additions: brachistochrone/cycloidal descent, sunflower/phyllotaxis, Voronoi partitions, Platonic solids, gyroid nodal approximation, projected Hopf fibers, trefoil tube and superformula shell.

Existing families include mycelial and slime-mold routing, honeycomb/hexagonal packing, leaf veins, vascular branching, river deltas, toroidal fields and other candidate forms. Each needs a task definition and appropriate comparator. The five Platonic solids are tetrahedron, cube, octahedron, dodecahedron and icosahedron; they form one registry family.

The June 19 frozen readiness baseline recorded 0 performance-ready, 3 legacy and 15 pending families. The October expansion preserves that history and makes no automatic promotions. The registry is a research map, not a list of validated inventions.

A new computation, with a narrow claim

Eight networks. Sixty-four nodes each. An ideal DC transport screen.

The October 4 run compares effective conductance and topological single-edge fault tolerance. Every network has normalized diameter 1, material volume 1 and resistivity 1; uniform conductor cross-section is set separately by each graph's total length. All 2,016 unordered terminal pairs are equally weighted.

Six graphs use the same terminal coordinates. The square grid is the reference. Mean conductance is in normalized $V/(\rho D^2)$; higher values mean greater conductance only within this declared model and comparison group.

Same-terminal network	Mean conductance	Single-edge deletions retaining full connectivity
Square grid	0.7643	100%
Loop-augmented tree	0.7091	100%
Triangulated square grid	0.6602	100%
Minimum spanning tree	0.2553	0%
Randomized tree control	0.1971	0%
Serpentine chain	0.1849	0%

The square grid leads this controlled subset in this one configuration. The result illustrates a tradeoff: more connections consume material, reducing each wire's cross-section under the fixed-volume policy. Loops can preserve connectivity after a broken edge. A zero full-connectivity rate does not mean no node pairs remain connected.

Honeycomb and sunflower networks were also computed, with mean conductance 1.2564 and 0.5235. Their terminal positions and boundary shapes differ, so they are **exploratory embeddings outside the controlled ranking**.

What the screen does not answer

No time, frequency, capacitance, inductance, impedance matching, attenuation, bandwidth, signal-to-noise, electromagnetic-wave or fluid-dynamics model is included. Contact resistance, finite conductor width, manufacture and post-failure conductance are also excluded.

The loop-augmented tree is a designed heuristic, not a validated fungal-growth model. One layout per family and one random-control seed are not multi-scenario validation. This is an internal synthetic screen; **the promotion gate remains false**. Stronger, longer-lasting real signals require a specified medium, frequency, dimensions, material, source/load and measured or validated simulated losses.

The historical research is broad - and includes failures

The June 18 canonical research audit preserves substantial earlier work. These are dated source-reported findings, not fresh reruns in this brief.

Historical evidence	Recorded scope/result	Interpretation
Master Universe V2	2,172 frozen public/market series; nine models/five families; 304 harmonic winners (14.0%)	Exploratory breadth under one 80/20 holdout; no universal edge established.
Hybrid Edge V7.1	80 untouched series, five outer folds; 11 screening-positive series; median MAE improvement 0%; zero robust claim gates	Adverse and neutral findings retained. Surrogate tests and multiple-comparison control were not run.
Phase-lock audit	1,764 candidate rows; 532 phase-lock-tagged; selected candidate ranks 557 by score and 631 by test Sharpe	Selected result underperformed its reported baseline. Candidate rows are not independent physical experiments.

A review of ten supplied screenshots found price-data views, rankings, projected-cost scenarios and an audit summary, but no visible geometry-comparison result. The pictured source audit prints zero measured sources and 57 adaptive-universe entries. Those entries must not be described as 57 measured feeds or validated geometries. The scope is the ten supplied images, not the entire historical photo archive.

From a candidate to a defensible finding

1. Define the geometry, material or algorithm and the specific question.
2. Freeze authorized inputs, comparator, budget, constraints and primary metric.
3. Use held-out scenarios, uncertainty analysis and controls appropriate to the search.
4. Preserve raw outputs, failures, environment, versions and reproducible commands.
5. Obtain independent or representative validation before operational claims.
6. Connect accepted technical results to buyer-authorized economics.

Release and claim gate: this is material prepared for public review. Publishing source or a demonstration does not certify the research, establish customer savings, or authorize production control. No operational geometry champion is asserted.

Start the review

[Domain: lumen-core.ai](#)

[Public repository](#)

[Reviewer starting point](#)

Source records: current config/geometry_championship_v1_registry.json; June 19 frozen geometry readiness scorecard; docs/LUMAUNIVERSE_RESEARCH_EVIDENCE_AUDIT_2026-06-18.md; October 4 synthetic-screen results and methodology; ten-image transcription audit. The dated registry, measured outputs and claim limits must travel together.